

6. Consider the following database for library management system

BOOK (Book_id, Title, Publisher_Name, Pub_Year)

BOOK_AUTHORS (#Book_id, Author_Name)

PUBLISHER (Name, Address, Phone)

BOOK_COPIES (#Book_id, #Branch_id, No-of_Copies)

BOOK_LENDING (#Book_id, #Branch_id, Card_No, Date_Out, Due_Date)

LIBRARY_BRANCH (Branch_id, Branch_Name, Address)

Write SQL queries to

- i. Retrieve details of all books in the library – id, title, name of publisher, authors, number of copies in each branch, etc.
- ii. Get the particulars of borrowers who have borrowed more than 3 books, but from Jan 2017 to Jun2017
- iii. Delete a book in BOOK table. Update the contents of other tables to reflect this data manipulation operation.
- iv. Create a view of all books and its number of copies that are currently available in the library.

```
CREATE TABLE BOOK (  
    Book_id INT PRIMARY KEY,  
    Title VARCHAR(50),  
    Publisher_Name VARCHAR(50),  
    Pub_Year INT  
);
```

```
CREATE TABLE PUBLISHER (  
    Name VARCHAR(50) PRIMARY KEY,  
    Address VARCHAR(100),  
    Phone VARCHAR(15)  
);
```

```
CREATE TABLE LIBRARY_BRANCH (  
    Branch_id INT PRIMARY KEY,  
    Branch_Name VARCHAR(50),  
    Address VARCHAR(100)  
);
```

```
CREATE TABLE BOOK_AUTHORS (  
    Book_id INT,  
    Author_Name VARCHAR(50),  
    PRIMARY KEY (Book_id, Author_Name),  
    FOREIGN KEY (Book_id) REFERENCES BOOK(Book_id)  
);
```

```
CREATE TABLE BOOK_COPIES (  
    Book_id INT,  
    Branch_id INT,  
    No-of_Copies INT  
);
```

```
Book_id INT,  
Branch_id INT,  
No_of_Copies INT,  
PRIMARY KEY (Book_id, Branch_id),  
FOREIGN KEY (Book_id) REFERENCES BOOK(Book_id),  
FOREIGN KEY (Branch_id) REFERENCES LIBRARY_BRANCH(Branch_id)  
);
```

```
CREATE TABLE BOOK_LENDING (  
Book_id INT,  
Branch_id INT,  
Card_No INT,  
Date_Out DATE,  
Due_Date DATE,  
PRIMARY KEY (Book_id, Branch_id, Card_No),  
FOREIGN KEY (Book_id) REFERENCES BOOK(Book_id),  
FOREIGN KEY (Branch_id) REFERENCES LIBRARY_BRANCH(Branch_id)  
);
```

```
INSERT INTO BOOK VALUES  
(101, 'DBMS', 'McGraw Hill', 2019),  
(102, 'Python Programming', 'Pearson', 2020),  
(103, 'Operating System', 'Oxford', 2018),  
(104, 'Computer Networks', 'McGraw Hill', 2021);
```

```
INSERT INTO PUBLISHER VALUES  
( 'McGraw Hill', 'New York', '9876543210'),  
( 'Pearson', 'London', '9876501234'),  
( 'Oxford', 'Oxford City', '9988776655');
```

```
INSERT INTO LIBRARY_BRANCH VALUES  
(1, 'Main Branch', 'Bangalore'),  
(2, 'City Branch', 'Mysore'),  
(3, 'North Branch', 'Mangalore');
```

```
INSERT INTO BOOK_AUTHORS VALUES  
(101, 'Navathe'),  
(102, 'Guido'),  
(103, 'Galvin'),  
(104, 'Tanenbaum');
```

```
INSERT INTO BOOK_COPIES VALUES  
(101, 1, 5), (101, 2, 3),
```

(102, 1, 4), (103, 3, 6),
(104, 2, 2);

```
INSERT INTO BOOK_LENDING VALUES
(101, 1, 1001, '2017-01-10', '2017-01-20'),
(102, 1, 1001, '2017-02-11', '2017-02-21'),
(103, 3, 1001, '2017-03-15', '2017-03-25'),
(104, 2, 1001, '2017-04-18', '2017-04-28'),
(101, 2, 1002, '2017-05-01', '2017-05-11'),
(102, 1, 1003, '2017-06-05', '2017-06-15');
```

-- i. Retrieve details of all books

```
SELECT B.Book_id, B.Title, B.Publisher_Name, BA.Author_Name, LB.Branch_Name,
BC.No_of_Copies
FROM BOOK B
JOIN BOOK_AUTHORS BA ON B.Book_id = BA.Book_id
JOIN BOOK_COPIES BC ON B.Book_id = BC.Book_id
JOIN LIBRARY_BRANCH LB ON BC.Branch_id = LB.Branch_id
ORDER BY B.Book_id;
```

-- ii. Borrowers with > 3 books between Jan 2017 and Jun 2017

```
SELECT Card_No, COUNT(Book_id) AS Total_Books
FROM BOOK_LENDING
WHERE Date_Out BETWEEN '2017-01-01' AND '2017-06-30'
GROUP BY Card_No
HAVING COUNT(Book_id) > 3;
```

-- iii. Delete a book (Cascading deletion)

```
DELETE FROM BOOK_LENDING WHERE Book_id = 101;
DELETE FROM BOOK_COPIES WHERE Book_id = 101;
DELETE FROM BOOK_AUTHORS WHERE Book_id = 101;
DELETE FROM BOOK WHERE Book_id = 101;
```

-- iv. Create view of currently available books

```
CREATE VIEW Available_Books AS
SELECT B.Title, BC.No_of_Copies
FROM BOOK B
JOIN BOOK_COPIES BC ON B.Book_id = BC.Book_id;
```

-- To test the view:

SELECT * FROM Available_Books;

